

A Deception Strategy Against Mobile Money Attacks Using Game Theory

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1 Related work

Recently, phishing attacks targeting Mobile Money services have drawn the attention of cybersecurity researchers due to their increasing frequency and impact on financial transactions. Numerous studies have been conducted to analyze these threats, covering various aspects such as fraud, social engineering attacks, and vulnerabilities in mobile infrastructures. Different approaches have been explored to identify and prevent these attacks.

In 2023, Alima Nzeket Njoya et al. used reinforcement learning to characterize phishing attacks targeting Mobile Money services. The evaluation of models (Q-learning, MDP, Deep Sarsa, A2C, DDPG, DQN) enabled the modeling of attacker-victim interactions and the identification of optimal attack sequences. Experiments conducted on real attack scenarios at Orange and MTN showed that Q-learning achieves more efficient attack sequences with less execution time. The authors further explored data availability limitations, highlighting the need for large-scale datasets for better model generalization[1].

Another study [2, 3], authors proposed a machine learning-based methodology for detecting smishing messages. Their approach involved ensemble techniques, utilizing multiple classification models such as Naïve Bayes, SVM, and Random Forest. The evaluation was conducted on the Spam Collection Dataset (Kaggle), achieving an accuracy of 98.9% in distinguishing smishing from legitimate messages. Their findings highlighted the effectiveness of ensemble learning in improving detection rates, though they acknowledged the need for multilingual evaluation to enhance generalizability.

This work[4], proposed by Franklin Tchakounte et al., introduces a game-theoretical model to anticipate email spear-phishing strategies. The study leverages Quantal Response Equilibrium (QRE) to predict an attacker's future intent based on past interactions between Intrusion Detection System (IDS) agents and malicious actors. The proposed approach enhances cybersecurity by strengthening phishing attack prevention. The authors achieved significant improvements in predictive accuracy, demonstrating the model's effectiveness in mitigating spear-phishing risks. A lot works have been done in 2020 as [5], proposed by Zakarya A. Alzamil, Abdulmalik Aljurayyad, Mohammed Almajally, Mohammed Abuhemid, Abdulmajeed Alsharafi, and Bader Alfreddi, introduces a hybrid phishing detection approach for mobile applications. The study leverages a combination of blacklist/whitelist techniques along with novel page detection and dummy data techniques to enhance phishing prevention. By analyzing webpage content and detecting suspicious behaviors, the proposed approach mitigates the limitations of existing anti-phishing solutions. The experimental study demonstrated superior detection accuracy, surpassing some commercial anti-phishing tools in identifying newly generated phishing links.

In 2017, [6] presents a vishing attack detection model for mobile users. The study investigates the psychological, technical, and information sensitivity factors that contribute to voice phishing (vishing) attacks. A rule-based model was developed to help mobile users detect social engineering attacks during phone conversations. The model incorporates decision-making rules based on emotional state, script clarity, caller identity verification, and sensitivity of requested information. The study findings highlight the increasing threat of vishing attacks in mobile banking and propose a proactive detection mechanism to mitigate user vulnerability.

Moreover, [7] proposed a novel system to prevent successful phishing attempts in text messages by incorporating game theory principles into modified Naïve Bayes and machine learning techniques. The results show that the proposed system achieved approximately 99% accuracy, while competitors had mean accuracy rates ranging from 91% to 86%.

Phiri et al.[8] propose an integrated model using both Natural Language Processing (NLP) and Machine Learning (ML) techniques to detect smishing attacks. Their approach utilizes Random Forest and

Naïve Bayes algorithms for classification, combined with NLP methods like Named Entity Recognition (NER) and Part-of-Speech (POS) tagging to extract relevant features from SMS messages. This combination aims to improve the accuracy and effectiveness of smishing detection compared to traditional methods.

This work [9] propose a game-theoretic framework for optimizing cybersecurity investment strategies in smart grids. This approach models the interaction between an attacker and a defender, each with their own objectives and constraints. The system takes into account the uncertain behavior of the attacker and the practical feasibility of the defender model. Same in [10] the paper surveys gamified solutions (2007-2019) designed to educate users about phishing. It analyzes these solutions based on their core objectives, game mechanics (points, badges, leaderboards), and learning processes. The research proposes three taxonomies to characterize gamified approaches to anti-phishing education and identifies areas for future development.

Table 1: Summary of research papers against phishing attcks

Paper	Authors	Year	Dataset	Methods	Acc.	Support	Limits	Attacks
[1]	Alima Nzeket Njoya, et al.	2023	attack scenarios (OM, MTN)	Reinforcement Learning	-	Prediction of mobile money phishing attacks	Lack of large-scale real data	Phishing
[11]	Adam Kavon Ghazi-Tehrani, et al.	2021	60 expert inter-views	Routine Activity Theory, qualitative analysis	-	Phishing persistence factors (motivation, skill gap)	Limited data	Phishing
[2]	Amey Dubey, et al.	2025	Spam Collection Dataset (Kaggle)	Machine Learning (Naïve Bayes, SVM, Random Forest)	98.9%	Smishing detection using ensemble learning	Limited dataset scope, need for multi-lingual evaluation	Smishing (SMS Phishing)
[3]	Iddi S. Mam-bina, et al.	2022	Swahili Smishing Dataset (31,962 messages)	Machine Learning (Random Forest, TF-IDF, ExtraTree)	99.86%	Detection of Smishing	Strong dependence on local data	Smishing (SMS Phishing)
[4]	Franklin Tchakounte, et al.	2020	Email-based data	Game Theory (QRE, Repeated Game)	-	Phishing strategy prediction	No real dataset, theoretical	Spear-Phishing (Email)
[5]	Zakarya A. Alzamil, et al.	2020	100 phishing URLs	Hybrid (Blacklist, Whitelist, Page Detection, Dummy Data)	-	Mobile phishing detection	Limited dataset, no real-time updates	Smishing, Phishing (Mobile)

Paper	Authors	Year	Dataset	Methods	Acc.	support	Limits	Attacks
[12]	Daniel Timko, et al.	2024	75 Community phishing messages	Experimental research design	75.8%	Anti-smishing technology study (NomoRobo)	Limited data	Smishing
[6]	Elijah Mwandata Maseno, et al.	2017	Mobile user interviews	Surveys, interviews	-	Vishing detection model	Small sample size	Voice phishing (Vishing)
[13]	Wenni Syafitri, et al.	2022	Literature review (2018-2021)	Systematic literature review	-	Social engineering prevention	-	Social engineering
[14]	Abid Jamil, et al.	2018	scenarios	Finite State Machine	-	Facebook phishing prevention	URL spoofing bypass	Social engineering
[15]	osin Ige, et al	2024	Scenarios	Literature review	-	Phishing detection techniques	Over-reliance on URLs	Phishing
[16]	Mousa, et al.	2022	45,300 URLs	Gated Recurrent Unit	98.30%	GRU phishing detection	Limited detection ability	Phishing URLs
[7]	Shambhavi, et al.	2024	5,574 SMS messages	Game Theory + Naïve Bayes	99%	MMS phishing detection	Limited real-world deployment	MMS Phishing
[8]	Katongo Ongani Phiri, et al.	2024	Real and synthetic SMS (Bemba, English)	NLP + ML (Random Forest, Naïve Bayes)	-	Smishing detection for mobile money	Small dataset, limited generalization	Smishing
[9]	Ige et al.	2024	-	Bayes, DL Comparison	-	-	Algorithmic Vulnerabilities	Email, Smishing, Vishing
[10]	Jafar et al.	2022	ISC2016	GRU Classifier System	98.30%	Better Accuracy	URL Feature Extraction	Phishing URLs
[17]	Zhiheng Xu, Jun Zhuang	2019	-	Sequential game with complete information	-	-	-	-
[18]	Chengwu Shao, Yan-Fu Li	2021	-	Graph Game	-	Protection Resources Allocation	-	sequential Cyber Attack
[19]	Azim, Abdallah	2024	Two-site game scenario	Game theory, QRE model	-	Analysis of Defender-Attacker	Only two sites modeled	Targeted site attacks

Paper	Authors	Year	Dataset	Methods	Acc.	support	Limits	Attacks
[20]	Devika Jay	2016	IEC-61850 VLAN	Deception, Game Theory	-	Decoy Allocation Strategy	Single-leader game	GOOSE, Replay, DoS
[21]	Azibek et al.	2021	Social Network	Game Theory Models	-	NE, Stackelberg	Budget Constraints	-
[22]	Guan Zhuang	2016	-	Sequential Game	-	Warm-Up Effects	-	-
[23]	Pawlick, Farhang, Zhu	2015	-	Signaling, FlipIt Games	-	Gestalt Equilibrium	Prior Commitment	-
Propose Model	-	-	scenarios base on dataset	Game theory+ LLM	-	Deception strategy to counteract mobile money phishing attcks	-	phishing

Main Objective of the Project

The main objective of this project is to design an intelligent deception system built around a **honeypot chatbot**, powered by a **Large Language Model (LLM)**. This chatbot will act as an **intermediary between the attacker and the intended victim** in the context of Mobile Money services. Its primary role is to **detect phishing attacks, deceive attackers, and gather valuable information** about their behavior.

The **game theory** framework will be used to **theoretically model the strategic interactions** between the chatbot (acting as the defender) and the attacker. This will help define optimal deception strategies that make it harder for attackers to succeed.

By providing **false but realistic information**, the chatbot will lure attackers into revealing their tactics, techniques, and procedures (TTPs). The data collected will be used to:

- continuously improve the chatbot’s defensive strategies,
- **detect and prevent future attacks more effectively,**
- **educate users** about new phishing techniques,
- and **strengthen the overall security of Mobile Money platforms.**

Specific Objectives

1. **Analyze phishing attacks** targeting Mobile Money users and organizations to understand the techniques used by attackers, such as social engineering, fake websites, or impersonation, to steal sensitive information like login credentials or access codes.
2. **Model the interaction between attackers and defenders** using game theory, specifically signaling games with evidence, to determine the most effective deception strategies to mislead attackers and protect users and transactions.
3. **Design and develop a deception chatbot**, based on a Large Language Model (LLM), capable of:
 - Simulating real Mobile Money users,

- Engaging attackers in realistic conversations,
 - Delivering false yet plausible information,
 - Collecting and analyzing attacker behavior (language, techniques, response patterns),
 - Generating realistic phishing scenarios for training and awareness.
4. **Evaluate the effectiveness of the deception strategy**, combining the chatbot and game-theoretic model, in a simulated environment. Measure its ability to reduce the success rate of phishing attacks and compare its performance with traditional Mobile Money security mechanisms.

2 System Architecture Overview

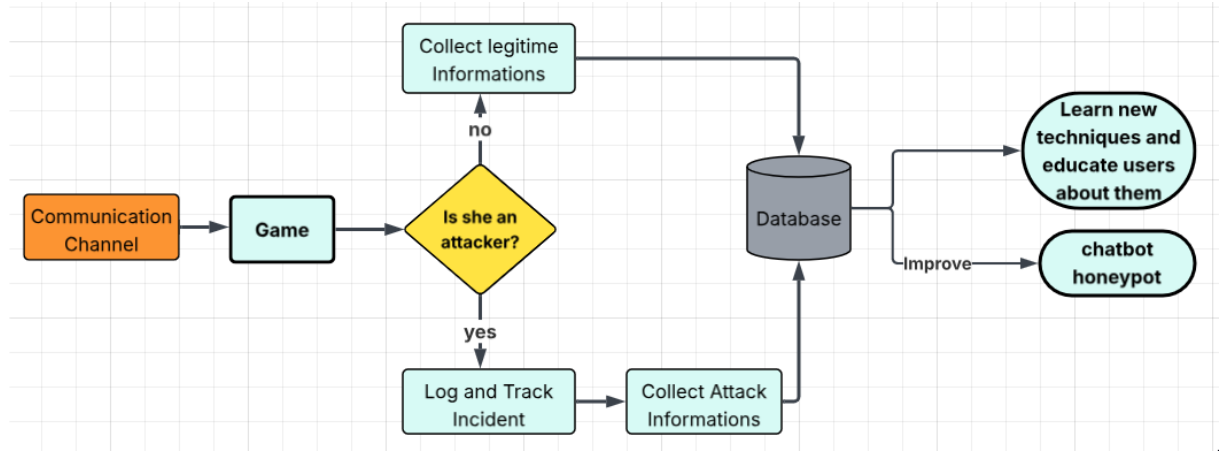


Figure 1: System Architecture

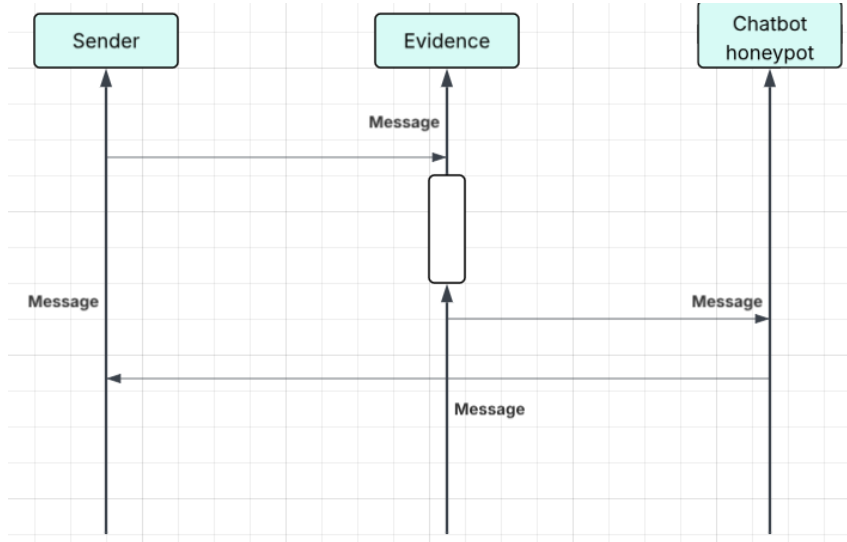


Figure 2: interaction game

3 Modeling a Signaling Game with Evidence in Mobile Money Systems

To model the strategic interaction between unknown mobile users and a defensive chatbot in a Mobile Money system, we adopt the signaling game with evidence framework introduced by Pawlick et al.

(2019)[24]. This game-theoretic approach captures how a system can detect and deceive malicious users (e.g., phishers or fraudsters) while maintaining uninterrupted service for legitimate users.

In this model, a sender (user) privately knows their type (honest or attacker) and sends a message. The receiver (defender chatbot) observes the message along with probabilistic evidence generated by a detection system and then chooses an appropriate action: to serve or to deceive.

4 Game Components

4.1 Players

- **Sender (S):** A mobile user, which can be of two types:
 - $\theta = 0$: A legitimate user.
 - $\theta = 1$: A malicious attacker (e.g., fraudster, phisher).
- **Receiver (R):** The defender chatbot, which observes the sender's message and some probabilistic evidence, and then chooses an action.

4.2 Messages and Evidence

- The sender chooses a message $m \in M = \{m_0, m_1\}$:
 - m_0 : Normal behavior (e.g., checking balance).
 - m_1 : Suspicious behavior (e.g., phishing attempt).
- Upon receiving the message, the system generates probabilistic **evidence** $e \in \{0, 1\}$:
 - $e = 1$: Suspicious activity is detected.
 - $e = 0$: No suspicious activity is detected.
- Detection probabilities:
$$\alpha = \mathbb{P}(e = 1 \mid \theta = 0) \quad (\text{False positive rate})$$

$$\beta = \mathbb{P}(e = 1 \mid \theta = 1) \quad (\text{True positive rate})$$

4.3 Actions

Based on the observed message m and evidence e , the receiver chooses an action $a \in A = \{a_0, a_1\}$:

- a_0 : Serve the user normally.
- a_1 : Engage deception (simulate a fake interface, monitor behavior).

Chatbot behavior policy:

- If $e = 0$, serve normally.
- If $e = 1$, activate deception.

5 Strategies and Beliefs

- **Sender's strategy:** $\sigma^S(m \mid \theta)$ is the probability that a sender of type θ sends message m .
- **Evidence generator:** $\lambda(e \mid \theta, m)$ is the probability that the system generates evidence e when a sender of type θ sends message m .
- **Receiver's strategy:** $\sigma^R(a \mid m, e)$ is the probability of taking action a given message m and evidence e .
- **Belief update:** After observing (m, e) , the receiver updates their belief about the sender's type using Bayes' rule:

$$\mu^R(\theta \mid m, e) = \frac{\lambda(e \mid \theta, m) p(\theta)}{\sum_{\tilde{\theta} \in \Theta} \lambda(e \mid \tilde{\theta}, m) p(\tilde{\theta})}$$

6 Utility Functions

6.1 Sender Utility

Let $u_S(\theta, m, a)$ denote the utility received by the sender when she is of type θ , sends message m , and the receiver chooses action a .

6.2 Receiver Utility

Let $u_R(\theta, m, a)$ be the utility received by the receiver for the same configuration. The receiver prefers to serve legitimate users and deceive attackers.

6.3 Assumptions

We adopt the following assumptions from the signaling game literature:

- Messages are *cheap talk*, i.e., they do not directly affect payoffs.
- The receiver gains more when he correctly classifies the sender's type.
- A malicious sender prefers to be misclassified (i.e., avoid deception).
- A legitimate user prefers normal service.

7 Expected Utilities

7.1 Sender's Expected Utility

$$\mathbb{U}^S(\sigma^S, \sigma^R | \theta) = \sum_{m, e, a} \sigma^S(m | \theta) \cdot \lambda(e | \theta, m) \cdot \sigma^R(a | m, e) \cdot u_S(\theta, m, a)$$

7.2 Receiver's Expected Utility

$$\mathbb{U}^R(\sigma^R | \theta, m, e) = \sum_a \sigma^R(a | m, e) \cdot u_R(\theta, m, a)$$

8 Equilibrium Concept

A **Perfect Bayesian Nash Equilibrium (PBNE)** in this game is a strategy profile (σ^S, σ^R) and a belief system μ^R such that:

- σ^S is a best response to σ^R for each type θ .
- σ^R is optimal given beliefs μ^R .
- Beliefs μ^R are consistent with Bayes' rule wherever possible.

8.1 Types of Equilibria

- **Pooling equilibrium:** Both types of senders send the same message.
- **Partially separating equilibrium:** Different types send different messages with certain probabilities, allowing the receiver to update beliefs.

9 System Goals

The defense system aims to:

- Accurately identify and deceive attackers.
- Avoid interrupting legitimate users.
- Gather behavioral data from suspicious interactions.

10 Game Flow

1. The sender observes their type θ and sends a message m .
2. Evidence e is generated according to $\lambda(e | \theta, m)$.
3. The receiver observes (m, e) , updates beliefs, and chooses action a .

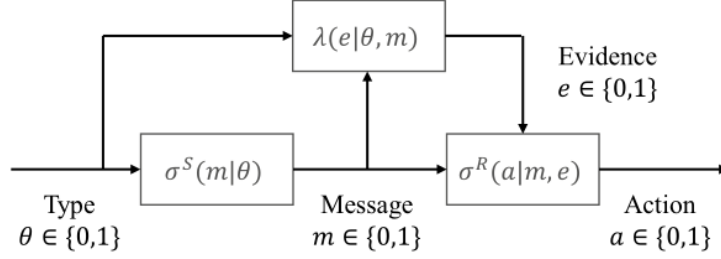


Figure 3: Signaling game with evidence in Mobile Money context [24]

11 Conclusion

This formal model provides a rigorous basis for designing defensive deception strategies in Mobile Money systems. It balances detection, deception, and service continuity, while enabling the study of attacker behavior and system robustness under uncertainty.

12 A Practical Case

In this practical scenario, we simulate the behavior of a deception-based chatbot installed locally on a Mobile Money user's phone. The chatbot is designed to scan incoming messages (such as SMS or instant messages), identify potential fraud attempts, and respond accordingly to either stall the attacker or continue the conversation with a legitimate contact.

Since the chatbot does not know in advance whether the sender is an attacker ($\theta = 1$) or a legitimate agent ($\theta = 0$), it relies on an internal detection system that assigns an **evidence flag** $e \in \{0, 1\}$ to each message:

- $e = 1$: the message is flagged as suspicious;
- $e = 0$: the message appears safe or normal.

Based on this evidence, the chatbot chooses one of two possible actions:

- A **deceptive response**, intended to trick, stall, or observe the sender without giving away any real data;
- A **normal (cautious) response**, meant to keep the conversation going and validate legitimacy if the message seems trustworthy.

12.1 The Social Engineering Cycle

Social engineering is a psychological manipulation technique used by cyber attackers to deceive individuals into revealing confidential information or performing harmful actions. Phishing, for example, is a common form of social engineering.

Social engineers do not act randomly. Instead, they follow a structured process to carry out their attacks effectively. The key stages of this cycle are presented in the figure below.



Figure 4: The social engineering attack cycle [1]



Figure 5: The phishing attack cycle[1]

Table 2: États S_1 à S_4 – Attaques par phishing

État	Actions typiques de l'attaquant	Réponse du Chatbot
S_1	- Impersonate a trusted third party - Collect information - Make contact through online social networks - Send message - Send message related to potential gain - Send email	D_1 : Active deception
S_2	- Activate the victim's emotions - Take the posture of benefactor - Conversation through social networks	D_1 : Active deception
S_3	- Activate balance consultation - Make false withdrawal - Recover cash - Send false message withdrawal - Push the secret code dialing - Use the identity of a relative of the victim - Activate transfer on behalf of the victim - Activate other transfer	D_1 : Active deception
S_4	- Recover secret code - Recover balance	D_1 : Active deception

États S_1 à S_4 – Attaques par phishing [1]

Les messages légitimes envoyés par des services comme MTN, Orange ou Airtel peuvent être regroupés en quatre grandes catégories fonctionnelles, correspondant aux sections S_5 à S_8 :

- **S_5 — Notifications de transaction** : messages confirmant une opération (envoi, retrait), indiquant le solde ou un reçu, ou annonçant un bonus officiel.
- **S_6 — Support client** : messages de réponse à une réclamation, de suivi de dossier ou d'assistance par canal sécurisé.
- **S_7 — Sécurité et prévention** : messages d'alerte, de rappel des bonnes pratiques et de sensibilisation contre la fraude.
- **S_8 — Paiements de services et offres** : confirmations de paiement de factures (Eneo, Camwater), d'abonnement (Canal+), ou de rechargement d'unités pour des services externes.

Table 3: États S_5 à S_8 – Messages légitimes des opérateurs

État	Actions typiques de l'agent légitime	Réponse du Chatbot
S_5	- Envoyer confirmation de transaction - Informer du solde - Envoyer un reçu électronique - Alerter sur une opération réussie - Notifier d'un bonus ou d'une offre (vérifiée)	D ₂ : Envoi de notification à l'utilisateur cible
S_6	- Répondre à une réclamation - Confirmer une enquête en cours - Proposer un lien officiel d'aide - Transmettre un identifiant de dossier - Répondre par un canal sécurisé	Envoi de notification à l'utilisateur cible
S_7	- Confirmer que jamais aucun agent ne demande un code - Alerter sur les comportements à éviter - Envoyer des conseils de sécurité - Vérifier que l'utilisateur est satisfait - Rappeler les moyens de contact officiels	D ₂ : Envoi de notification à l'utilisateur cible
S_8	- Notification de paiement de facture (Eneo, Camwater, etc.) - Confirmation d'abonnement à Canal+, StarTimes, etc. - Reçus de paiements scolaires ou administratifs - Notifications de rechargement d'unités (crédit énergie, eau) - Confirmation de paiement via Mobile Money pour un service externe	D ₂ : Envoi de notification à l'utilisateur cible

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